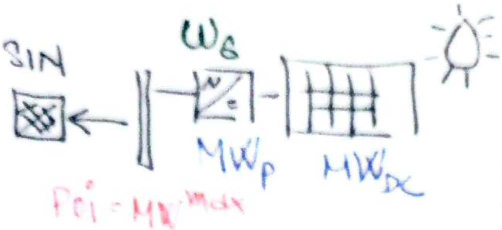
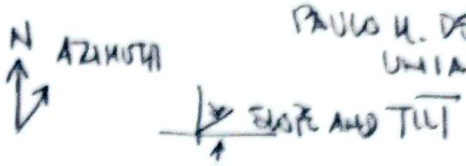
OPTIMIZACIÓN AGPE + PV + BESS
CURSO EDCO CLASE 2 MODULO 1

MODELO DE NEGOCIO UTILITY - SCALE

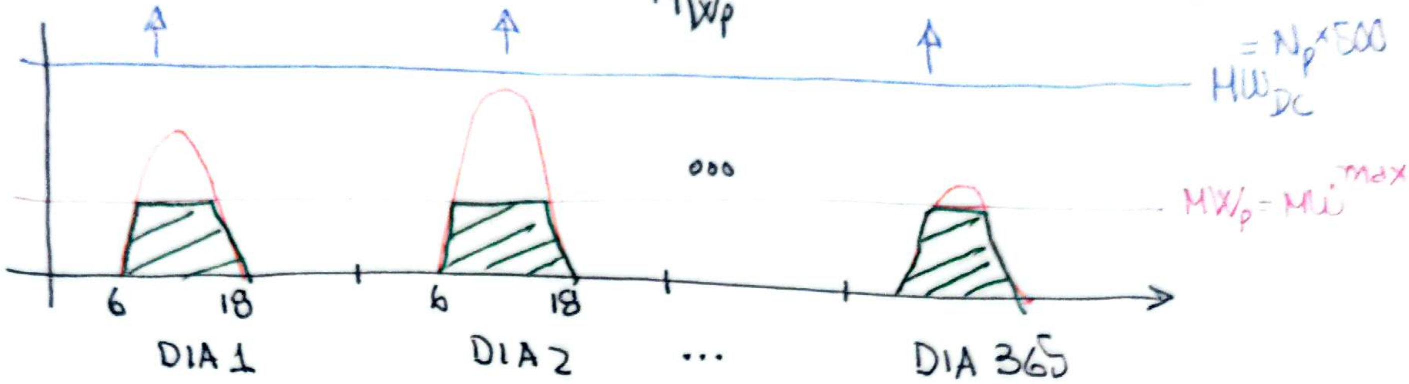
PAULO M. DE OLIVEIRA
UNIANDES
2026





PVGIS 5.3 19 year DATABASE 2005 - 2023

$$I_{LR} = \frac{MW_{DC}}{MW_p}$$



ENERGÍA A VENDER
AREA BAJO LA CURVA

$$\omega_S \triangleq \sum_{t=1}^{T=8760h} P_{PV}^t \qquad P_{PV}^t < MW^{max} = MW_p \quad \forall t = 1, \dots, 8760$$

MODALIDADES:

1.- MERCADO SPOT (BOLSA) [PAY-AS-CLEAR] $\rightarrow E_S = \sum_{t=1}^{8760} P_{PV}^t \cdot \lambda^t$
MARGINALISTA
VENTAS
SPOT PRICE DAY AHEAD

2.- SUBASTAS [PAY-AS-BID]
3.- CONTRATOS [PPA]
} PRECIO ÚNICO λ

BREAK-EVEN PRICE

$i \triangleq$ WACC, % ANUAL
$e \triangleq$ INFLACIÓN ANUAL
$n \triangleq$ VIDA DEL PROYECTO, AÑOS

EMBER: $i = 7.7\%$ $n = 10$ yrs
$e = 2.5\%$

$$LCOE = \frac{CAPEX + \dfrac{OPEX}{CRF_e}}{\omega_S / CRF}$$

$$CRF = \frac{i(i+1)^n}{(i+1)^n - 1}$$

$$CRF_e = \frac{i - e}{1 - \left[\frac{1+e}{1+i}\right]^n}$$

$$CAPEX = MW_{DC} \cdot K_{DC} + MW_p \cdot K_p + BoP$$

$K_{DC} \rightarrow 0.4 \frac{USD}{W}$ $\quad K_p \rightarrow 0.05 \frac{USD}{W}$

$$OPEX = MW_{DC} \, K_{OM}$$

$K_{OM} \rightarrow 0.0125 \frac{USD}{W}$

$$NPV \triangleq \frac{E_S - OPEX}{CRF_e} - CAPEX \geq 0$$

$$NPV = 0 = -CAPEX + \frac{(E_S - OPEX)}{\dfrac{IRR - e}{1 - \left[\frac{1+e}{1+IRR}\right]^n}}$$

$\rightarrow$ INTERNAL RATE OF RETURN

$$NPV = 0 = -CAPEX + \frac{\frac{(E_s - OPEX)}{i-e}}{\left(1-\left(\frac{1+e}{1+i}\right)^{PBT}\right)} \rightarrow PBT$$

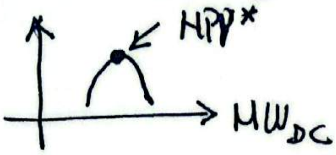
PAY BACK time

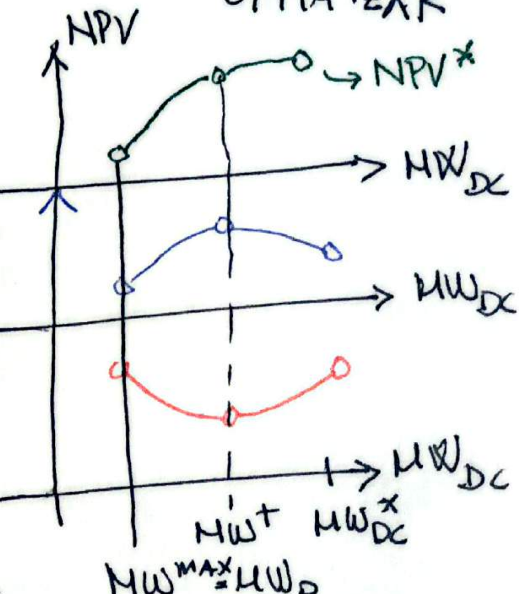
OPTIMIZATION PROBLEM

$$\max_{MW_{DC}} NPV \quad \text{subject to} \quad \frac{MW_{DC}}{ILR} = MW_1^{max}$$

UNA SOLA VARIABLE DE DECISION → NO ES NECESARIO OPTIMIZAR

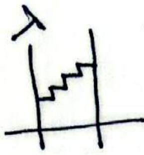
SIMULACIÓN MEDIANTE PVGIS 5.3 PARA λ^t, λ



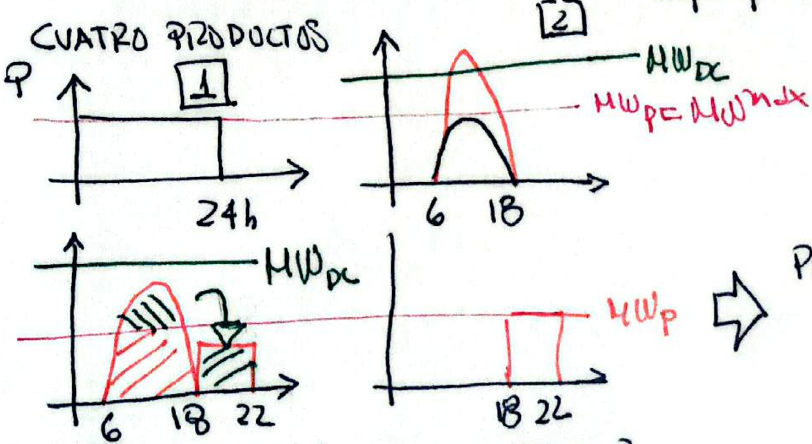


$MW^+ \equiv$ SOLUCIÓN EFICIENTE

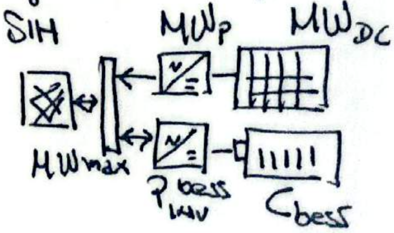
¿CON QUE λ VOY A LA SUBASTA?



CUATRO PRODUCTOS



SIN

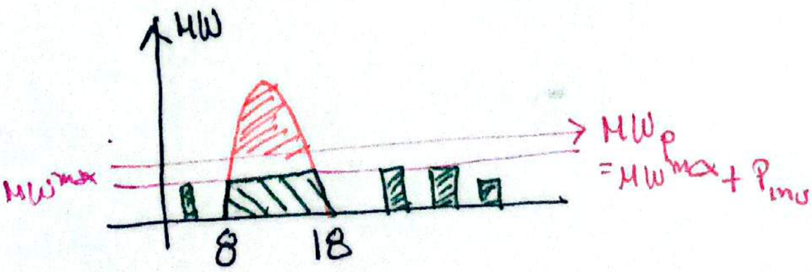


⇒ PROBLEMA DE OPTIMIZACIÓN 3 VARIABLES:

- MW_{DC}
- POTENCIA INVERSOR BES
- CAPACIDAD BES

$$MW_p = MW^{max} + P^{bess}_{inv}$$

¿MERCADO SPOT?



PROBLEMA DE OPTIMIZACIÓN

3 INCÓGNITAS: MW_{DC} MW_p C

GESTIÓN BATERIA = 8760 x 2

S_oC → PROGRAMACIÓN BES